Vector Spaces V: Bases, Dimension and Polynomial Spaces

Problem sheet for the Day 6 class · CSIR NET / GATE previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. CSIR NET Dec 2016 · MCQ

Which of the following subsets of $\mathbb{R}^4$ is a basis of $\mathbb{R}^4$?

$B_1 = \{(1,0,0,0), (1,1,0,0), (1,1,1,0), (1,1,1,1)\}$,
$B_2 = \{(1,0,0,0), (1,2,0,0), (1,2,3,0), (1,2,3,4)\}$,
$B_3 = \{(1,2,0,0), (0,0,1,1), (2,1,0,0), (-5,5,0,0)\}$.

(A) B_1 and B_2 but not B_3
(B) B_1, B_2 and B_3
(C) B_1 and B_3 but not B_2
(D) Only B_1

2. GATE 2012 · MCQ

Let $V = \mathbb{C}^2$ be the vector space over the field of complex numbers and $B = \{(1,i), (i,1)\}$ be a given ordered basis of V. Then for which of the following, $B^* = \{f_1, f_2\}$ is a dual basis of B over $\mathbb{C}$?

(A) $f_1(z_1,z_2) = \frac{1}{2}(z_1 - iz_2),\ f_2(z_1,z_2) = \frac{1}{2}(z_1 + iz_2)$
(B) $f_1(z_1,z_2) = \frac{1}{2}(z_1 + iz_2),\ f_2(z_1,z_2) = \frac{1}{2}(iz_1 + z_2)$
(C) $f_1(z_1,z_2) = \frac{1}{2}(z_1 - iz_2),\ f_2(z_1,z_2) = \frac{1}{2}(-iz_1 + z_2)$
(D) $f_1(z_1,z_2) = \frac{1}{2}(z_1 + iz_2),\ f_2(z_1,z_2) = \frac{1}{2}(-iz_1 - z_2)$

3. GATE 2013 · NAT

Let V be the real vector space of all polynomials in one variable with real coefficients and having degree at most 20. Define the subspaces
$W_1 = \{p \in V : p(1) = 0,\ p(\frac{1}{2}) = 0,\ p(5) = 0,\ p(7) = 0\}$ and
$W_2 = \{p \in V : p(\frac{1}{2}) = 0,\ p(3) = 0,\ p(4) = 0,\ p(7) = 0\}$. Then the dimension of $W_1 \cap W_2$ is _____.

4. GATE 2013 · MCQ

Let M be the space of all 4×3 matrices with entries in the finite field of three elements. Then the number of matrices of rank three in M is

(A) $(3^4 - 3)(3^4 - 3^2)(3^4 - 3^3)$
(B) $(3^4 - 1)(3^4 - 2)(3^4 - 3)$
(C) $(3^4 - 1)(3^4 - 3)(3^4 - 3^2)$
(D) $3^4(3^4 - 1)(3^4 - 2)$

5. CSIR NET June 2016 · MCQ

Consider the real vector space V of polynomials of degree less than or equal to d. For $p \in V$ define $\|p\|_k = \max\{|p(0)|, |p^{(1)}(0)|, \ldots, |p^{(k)}(0)|\}$, where $p^{(i)}(0)$ is the i th derivative of p evaluated at 0. Then $\|p\|_k$ defines a norm on V if and only if

(A) $k \geq d - 1$
(B) $k < d$
(C) $k \geq d$
(D) $k < d - 1$

6. CSIR NET June 2016 MCQ Let V be the vector space of all real polynomials of degree ≤ 10. Let $Tp(x) = p'(x)$ for $p \in V$ be a linear transformation from V to V. Consider the basis $\{1, x, x^2, \ldots, x^{10}\}$ of V. Let A be the matrix of T with respect to this basis. Then

(A) $\mathbf{Trace}\, A = 1$

(B) $\det A = 0$

(C) there is no $m \in \mathbb{N}$ such that $A^m = 0$

(D) A has a nonzero eigenvalue

7. CSIR NET June 2016 MSQ Let V be the vector space of all complex polynomials p with $\deg p \leq n$. Let $T : V \to V$ be the map $(Tp)(x) = p'(1)$, $x \in \mathbb{C}$. Which of the following are correct?

(A) $\dim \mathrm{Ker}\, T = n$

(B) $\dim \mathbf{range}\, T = 1$

(C) $\dim \mathrm{Ker}\, T = 1$

(D) $\dim \mathbf{range}\, T = n + 1$

8. CSIR NET Dec 2017 MSQ Let V be the vector space over $\mathbb{C}$ of all polynomials in a variable X of degree at most 3. Let $D : V \to V$ be the linear operator given by differentiation with respect to X. Let A be the matrix of D with respect to some basis for V. Which of the following are true?

(A) A is a nilpotent matrix

(B) A is a diagonalizable matrix

(C) the rank of A is 2

(D) the Jordan canonical form of A is $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

9. CSIR NET Dec 2015 MSQ Let V be a finite dimensional vector space over $\mathbb{R}$. Let $T : V \to V$ be a linear transformation such that $\mathbf{rank}(T^2) = \mathbf{rank}(T)$. Then,

(A) $\mathbf{Kernel}(T^2) = \mathbf{Kernel}(T)$

(B) $\mathbf{Range}(T^2) = \mathbf{Range}(T)$

(C) $\mathbf{Kernel}(T) \cap \mathbf{Range}(T) = \{0\}$

(D) $\mathbf{Kernel}(T^2) \cap \mathbf{Range}(T^2) = \{0\}$

10. CSIR NET Dec 2015 MSQ Let V be the vector space of polynomials over $\mathbb{R}$ of degree less than or equal to n. For $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ in V, define a linear transformation $T : V \to V$ by $(Tp)(x) = a_n + a_{n-1} x + \cdots + a_0 x^n$. Then

(A) T is one to one

(B) T is onto

(C) T is invertible

(D) $\det T = \pm 1$